

Same Energy Mix, Two Economic Systems

What happens to the US wind + PV + solar-thermal + CSP + sand-battery mix when the RCED planning institutions are removed and the current market economy decides what gets built

Key finding: the technology is identical in both cases — the climate outcome is set by the institutions, not the hardware.

Executive Summary

A previous study showed that, under the planned RCED architecture (Layers 12–14), a United States system built on wind, solar PV, solar thermal, CSP and a sand battery reaches 100% reliability with near-zero emissions. This report keeps the **identical generation mix and demand** but removes the RCED's planning instruments and lets the **current US market economy** govern what is built and how gaps are filled. The flexible resource becomes natural gas, as it is today.

The result is stark. Forcing the same enormous renewable fleet yields 99% renewable electricity but wastes **32.8%** of it through curtailment, leaves heat 27% fossil, and still emits **286 Mt CO2/yr**. But a profit-driven market would never build that fleet, because the wasted third makes the marginal investment unprofitable. Sized to a merchant-rational level, the same mix delivers only 84% renewable electricity, leaves heat **70%** fossil, and emits **1011 Mt CO2/yr** — roughly 3.5× the planned case and around 200× the RCED outcome.

1. Method: what "removing the RCED model" means

The physical system is unchanged: the same installed capacities (PV 2,450 GW, wind 1,450 GW, CSP 400 GW, solar thermal 450 GW) and the same final demand of ~11,400 TWh/yr are simulated hour by hour. Only the institutional rules change, to reflect how the US market actually behaves:

- **No seasonal storage.** An asset that charges and discharges only once a year earns negligible revenue per unit of power, so the market does not build inter-seasonal sand or hydrogen stores. It builds 4-hour batteries, which do nothing for the winter deficit.
- **No district-heat networks.** The US has almost no urban heat grids, so the sand battery — the centerpiece of the RCED heat strategy — cannot deliver heat to buildings. Its role collapses to a small on-site industrial buffer.
- **A fragmented grid.** Without a planned national HVDC mesh, the three US interconnections provide less spatial smoothing, so the continental-scale benefit is reduced.
- **Natural gas as the flexible resource.** Gas CCGTs and furnaces fill every electricity and heat gap, exactly as in today's system, with residual CO2 accounted for.

2. Results

Indicator	RCED (planned)	Market — same fleet	Market — profitable fleet
Renewable share of electricity	100%	99%	84%
Renewable curtailment	3.8%	32.8%	7.5%
Heat supplied by fossil gas	0%	27%	70%

Gas for electricity (TWh)	0	41	834
Residual CO2 (Mt/yr)	~5	286	1011

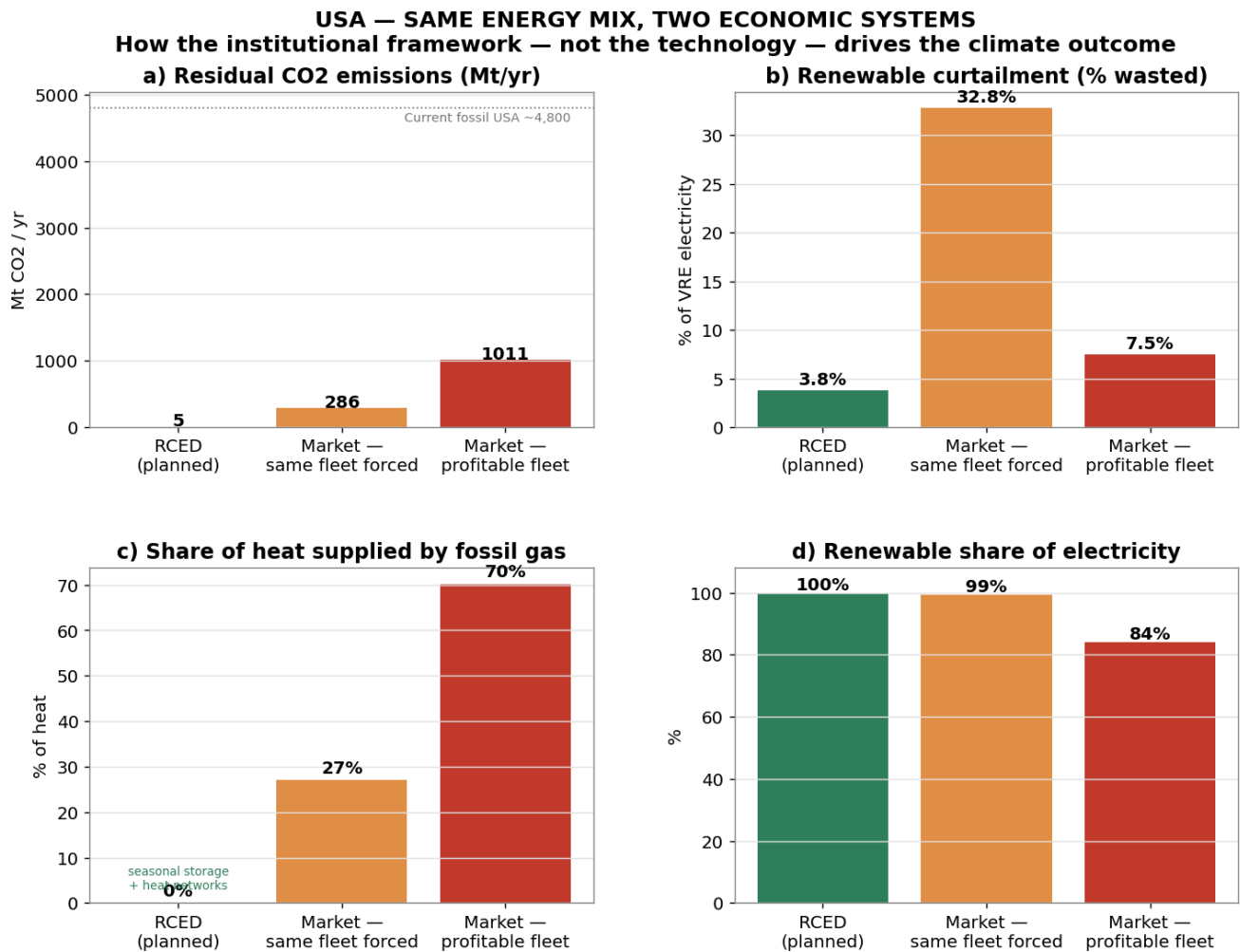


Figure 1 — The same mix under two systems: (a) residual CO2 emissions; (b) renewable curtailment; (c) share of heat from fossil gas; (d) renewable share of electricity. Green = planned (RCED); orange = market with the same fleet forced; red = market sized to be profitable.

3. Analysis

3.1 Seasonal storage is uneconomic in a market.

The decisive gap between the systems is the winter. The RCED stores summer surplus as heat in the sand battery and as firm reserve for electricity, then releases it months later. In a market, a store that completes one cycle per year cannot recover its capital — its revenue per megawatt is a fraction of a daily-cycling battery's. So the market builds short-duration batteries that smooth the day but are useless against the seasonal deficit, and it leans on gas instead. This single economic fact explains most of the emissions gap.

3.2 Curtailment caps the renewable build.

Forcing the full fleet produces 99% renewable electricity, but a third of it is spilled. For a generator, spilled energy is unpaid energy: as penetration rises, additional solar and wind capture ever-lower prices (cannibalization), and the marginal project stops being financeable. A profit-seeking system therefore halts at a far smaller fleet — here about 55% of the planned capacity — and the remaining demand is met by gas. The renewable share of electricity falls from 99% to 84%, and the heat sector,

which depended on cheap surplus electricity to run heat pumps, reverts overwhelmingly to gas.

3.3 No heat networks, no electrified heat.

Even where surplus electricity exists, the absence of district-heating infrastructure means the sand battery cannot serve buildings. Heat decarbonization in the market case rests entirely on individual heat pumps running when the grid happens to be clean — which, with a smaller fleet and no seasonal storage, is not often enough in winter. The heat sector is where the institutional difference bites hardest: 0% gas in the planned system versus 70% in the market-rational one.

4. Limitations

- Gas is treated as an unlimited flexible resource; real peaking constraints and price dynamics are simplified.
- The merchant fleet is sized by an ~8% curtailment heuristic rather than a full hourly price-based investment model.
- Demand is held identical to isolate the institutional effect; a real market would also electrify heat and transport more slowly, which would *worsen* the CO2 result further.
- Synthetic weather, single year, single national node (no inter-regional congestion).

5. Conclusion

The same wind, sun, turbines and panels can decarbonize the United States almost completely — or leave it emitting on the order of a gigatonne of CO2 a year. The difference is not technological; it is institutional. The RCED's value lies precisely in the assets and coordination that a market does not finance on its own: inter-seasonal storage, urban heat networks, a planned national grid, and a deliberate acceptance of overbuild. Strip those away and the identical hardware delivers a partial, gas-dependent transition, because the price signal rewards short-duration flexibility and penalizes the seasonal investment that winter demands.

The practical implication is not that markets cannot decarbonize, but that they will not reach the last and hardest tranche — winter heat and the seasonal electricity deficit — without policy that pays for what hourly prices do not: long-duration storage, heat infrastructure, and transmission. Whether that policy takes the comprehensive form of the RCED or a narrower set of mandates and capacity payments, the simulation makes the same point quantitatively: the institutional framework, far more than the technology, decides how much of the climate problem actually gets solved.